

Remembering Things Makes Chatbots Sound Smarter, but Less Trustworthy - A Pilot Study

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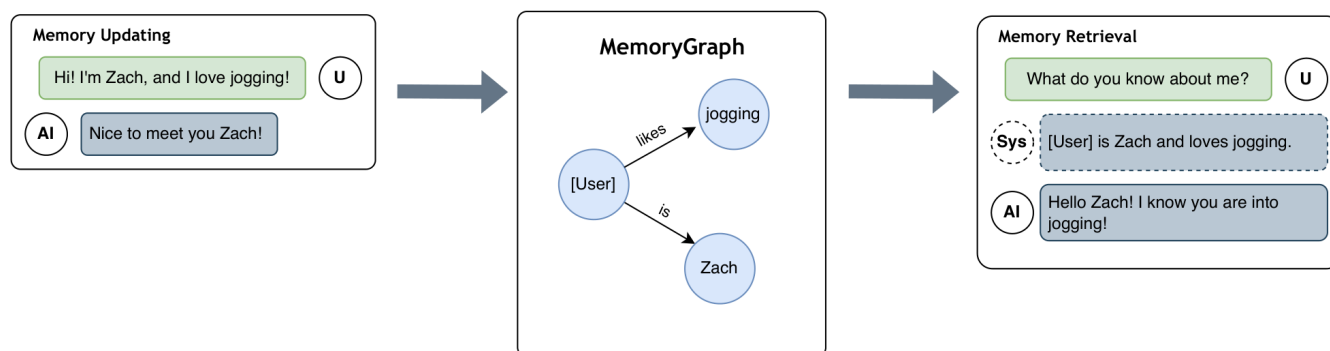


Figure 1: Overview of MemoryGraph. The system extracts triples from the conversation and stores them in the knowledge graph. Upon new queries that require memory retrieval, the system explores the knowledge graph and generates a query-focused summary for generating responses.

Abstract

Memory is a key aspect of human interaction, allowing us to remember facts about our interlocutors and personalise the way we interact. This pilot study investigates how a semantic long-term memory affects user assessments of chatbot likeability, perceived intelligence, and perceived safety, addressing limitations in Large Language Model (LLM) memory. We introduce MemoryGraph, a knowledge graph-based memory system for LLMs that enables visual inspection of memories by users. A user study compared interactions with a chatbot under three conditions: no memory, memory alone, and memory with visualisation. User ratings for likeability, intelligence, and safety were recorded. Preliminary findings show that adding a memory without visualisation reduced positive assessments compared to the baseline, whereas combining memory with visualisations improved these ratings. This tentatively suggests that the observability of memory functions influences key user perceptions of memory-augmented conversational AI, although more research is needed to confirm these initial results.

CCS Concepts

• Computing methodologies → Natural language processing; Semantic networks.

Keywords

memory-augmented conversational AI, knowledge graph, semantic long-term memory, user study

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1 Introduction

Users of large language models (LLMs) often anthropomorphise them by attributing cognitive abilities beyond their demonstrated capacities [3, 4, 9]. This tendency stems from the models' proficiency in generating human-like text, handling complex language, and accessing broad knowledge bases. Their capacity for extended conversation, supported by large context windows, also contributes to this perception. However, a significant limitation is their restricted use of long-term memory [12]. Most current commercial models typically do not retain information across separate interactions by default. Some specific models such as Google Gemini and OpenAI's ChatGPT have recently introduced some notion of session-free memory, but it is somewhat limited to free text.

This pilot study aims to investigate the following research question: **Does the presence of long-term dynamic semantic memory in conversational AI agents affect the trust of its users?** In order to answer this question, we introduce MemoryGraph, a system that couples an LLM with a knowledge graph-powered semantic memory. We aim to measure the effect of this memory component on user assessments of three criteria: chatbot **likeability**, **perceived intelligence**, and **perceived safety**. While preliminary, the findings provide information regarding factors influencing perceptions of LLM intelligence and the potential of dynamic knowledge graphs in improving the design of human-facing conversational AI systems.

In section 2 we go over the main paradigms for LLM-focused memory usage and representation. In section 3, we detail how the implementation of the memory mechanism using knowledge graphs. In section 4, we explain the design of our preliminary user study on the effect of this semantic memory on user perception. Finally, in sections 5 and 6, we present and discuss the results of our preliminary study, and use those results to provide recommendations for future work.

2 Related Work

Previous research published in CUI has shown that users want the chatbot to remember past interactions [17], has proposed **memory intents as a way to imbue agents with conversational memory** [12], **and to use knowledge graphs to harness dynamic domain knowledge** [8]; it has also stressed the utility of memory if LLMs are to successfully play the role of a management coach [6]. Integrating memory into chatbots probably has wide-ranging utility in areas such as education [14], and health settings such as dementia care [16].

Previous studies have explored the use of text-based external memory base to power LLMs with long-term memory capabilities.

Unstructured memories. While **text-based memory mechanisms, such as those proposed by Zhong et al. [21] and Lu et al. [11], provide efficient methods for summarising and retrieving past conversations, they may struggle with memory recall when the information needed is not semantically similar to the users' query, which hinders their abilities to perform complex reasoning.** Our proposed **knowledge graph-based memory mechanism addresses this by representing information as entities interconnected with edges.** Similarly, Wang et al. [19] proposed a **memory mechanism that also uses the summary of past conversations to improve LLMs with long-term memories.** Unlike the previous two memory mechanisms that retrieve related conversation summaries to generate relevant responses, their system introduces a **memory controller that uses an LLM to determine whether it is necessary to retrieve summaries.** Like other studies mentioned, the memory mechanism introduced by Zheng et al. [20] is also designed to work with open-weight models such as Gemma 2 [18] or Llama 3.1 [5] and closed-source models such as ChatGPT. Their system was designed for non-playable characters (NPCs) in video games, allowing them to act as humans while keeping important information in the memory base and discarding others. Their system has a selective forgetting mechanism that mimics human memory processes to remove unused memories

from the memory base and improve further retrieving process. Despite this, they found that their forgetting mechanism can lead to memory loss and cause a decrease in memory retrieval accuracy, which means that the **trade-off between forgetting unimportant details and preserving critical information could hinder the ability of their NPCs to recall relevant past interactions,** especially in some complex dialogue contexts.

Packer et al. [15] proposed a plug-and-play system called MemGPT, which allows LLM trained on a tool calling dataset to write or retrieve from the memory base. The retrieved memories are stored in a fixed-size window within the context window. Although allowing **the LLM to decide when to recall memories and perform chained tool calling** (i.e., calling tools with different keywords or parameters until it decides to send a message to the user) is more flexible, it introduces potential issues such as requiring the LLM to generate a large number of tokens to complete the process, which could result in increased latency and higher computational costs. Additionally, this mechanism may be difficult to implement in LLMs that are not specifically trained or fine-tuned on a tool-calling dataset, making it harder to deploy with different LLMs.

Structured memories. Liu et al. [10] introduced a **new memory mechanism based on extracted triples (subject, relation, object), where these triples are defined as thoughts derived from Question-Answer pairs between the user and the chatbot.** This has several limitations. Instead of being stored in a graph database, the triples are converted directly into text, making it difficult to perform any form of graph-based exploration. Furthermore, the triple extraction process relies on only individual message pairs and ignores the broader conversation history. This limitation hinders coreference resolution as important contextual information from earlier conversations is ignored. Packer et al. [15] proposed a long-term memory mechanism called MemGPT. Using LLM trained on tool calling datasets, it enables memory tools to be activated. In addition to the mentioned tools, it also includes a key-value pair store for additional memory capabilities.

Hu et al. [7] explored the use of SQL databases as **external memories for LLMs.** The system first breaks down the instruction into several steps and then uses Chain-of-Thought to allow the LLM to generate new SQL queries based on the response of all previous steps. This approach makes it flexible for the schema and is capable of handling exceptions (e.g., some data it tries to query or delete might be missing).

Modarressi et al. [13] implemented a long-term memory mechanism using a fine-tuned LLM to extract triples from user input or generate triplets that have missing elements that serve as a query. This research explored the approach to using knowledge graph-based external memory storage. **One major drawback of this approach is that it relies entirely on the exact match for names, making it impossible to perform a fuzzy search for entities or relationships.** Similar works, such as AutoKG [2], have explored the **construction of knowledge graphs using LLM for entity linking.**

3 Semantic Memory Mechanism

Knowledge graphs allow us to structure information as nodes (concepts) and edges (relationships), and enable chatbots to navigate and reason about complex concepts. Our system is implemented on

top of a Neo4J graph database¹, which handles the underlying operations of creating, updating, and deleting entities and relationships. In this section, we describe two main mechanisms of the system: *updating* and *retrieving* memories.

3.1 Memory Updating

The memory updating process involves representing the current conversation as a graph of related entities before merging this graph into the knowledge graph representing the memory of the model, as illustrated in Figure 2. It comprises several steps, detailed in this section.

3.1.1 Sub-graph Construction.

Entity extraction and typing. This step focuses on identifying and **extracting entities from the user’s current conversation, to resolve coreferences**. The process identifies key details for each entity, such as its name, type, and description within the message. As illustrated in Figure 3, the extracted entities are highlighted and typed, such as “Person” or “Hobby”. All entities are output in a single list in a standardised JSON format. This functionality is implemented using an LLM agent designed to identify all entities within the current message, including those whose references have been resolved from previous conversations. Additionally, a special “[User]” entity is introduced to represent the user, which acts as the default entry point for the knowledge graph in future retrieval steps.

Relationship extraction. This step focuses on **identifying the relationships between the entities extracted in the previous stage**. Using the context of the user’s message and the conversation history, an LLM agent determines how these entities are related to each other, such as “lives in”, “loves”, or “is the friend of”. Figure 4 shows the pipeline for extracting relationships, resulting in a sub-knowledge graph.

Defining missing entities. To make sure every entity that appeared in the extracted relationships is properly extracted and defined, we use a separate LLM agent designed to define missing entities. In most cases, the “[User]” Entity is likely missing in the initial entity extraction step.

3.1.2 Knowledge Graph Integration. Figure 5 shows the pipeline for integrating an extracted subgraph into the main knowledge graph. The knowledge graph integration aims to update the main knowledge graph without creating duplicate entities or relationships. In the entity-linking step, the system first tries to find the same entity in the main knowledge before creating a new one. For relationships in the knowledge subgraph, the system searches semantically similar relationships before creating new ones.

After linking both entities and relationships, an LLM agent generates a summary for the entities in the sub-knowledge graph. The description of the entity is updated using the generated summary, and the corresponding sentence embeddings are updated for future semantic entity searches.

Entity linking. The entity linking step involves mapping the extracted entities from the sub-knowledge graph construction phase to their counterparts in the existing knowledge graph, as shown

in Figure 6. **The system will first try to find entities in the sub-knowledge graph that are the same as those in the main knowledge graph. If there are entities with the same name and type, the entities are directly linked. For entities that are not directly linked, the system first tries to find entities that are semantically similar. This is done by retrieving k entities that are most semantically similar to the unlinked entities.**

To determine if the retrieved entities are the same as the unlinked entities, an LLM agent prompted with instructions and a few-shot examples assess whether the entities are identical. Properties such as name, description, and type are used to perform fuzzy matching. All entities that have a corresponding counterpart in the main knowledge graph are linked. For entities that cannot be linked, the system will create them in the main knowledge graph.

Relationship linking. In this step, the system retrieves potential relationships from the existing knowledge graph that are similar to those extracted in the previous phase, as illustrated in Figure 6. **The similarity between relationships is determined by calculating the cosine similarities of their description embeddings (with a threshold arbitrarily set to 0.9).** An LLM agent is then used to assess whether these relationships are identical. This ensures that semantically equivalent relationships, such as “is a student of” and “studies at”, are recognised as similar. If no similar relationships are found, the system will create a new relationship. Identifying similar relationships helps prevent the creation of redundant relationships between entities.

Entity relationships summarisation. To enhance future entity linking and reasoning, the system generates a summarised description of all bi-directional relationships associated with the updated entities (illustrated in Figure 7). The process starts by retrieving all relationships connected to the entity that has been updated. In certain cases, the LLM agent responsible for generating summaries will confuse the incoming and outgoing relationships. To solve this, the system will first reverse all incoming relationships (e.g., “supervised by” becomes “supervises”) to create a more accurate description. Then, the LLM agent will generate a paragraph describing all 1-hop relationships to the entity. The summaries generated from entities provide an overview of the entity within the knowledge graph.

3.2 Memory Retrieval

Memory retrieval (illustrated in Figure 8) processes a *query* alongside *conversation_history* to retrieve relevant information and generate a summary. The key steps are:

- (1) **Query rewriting:** The initial *query* is contextually refined using the *conversation_history* via an LLM.
- (2) **Entry entity determination:** Entities are extracted from the refined query. A primary *entry_entity* is identified from these candidates and the conversation history. If no entity is found, a default is used.
- (3) **Chain-of-Thought reasoning:** This step iteratively constructs a *reasoning_path* starting from the *entry_entity*:
 - 3.1 Relationships linked to the current entity are retrieved.
 - 3.2 An LLM prunes these relationships for query relevance.
 - 3.3 An LLM selects the next relationship for the path.

¹<https://neo4j.com>

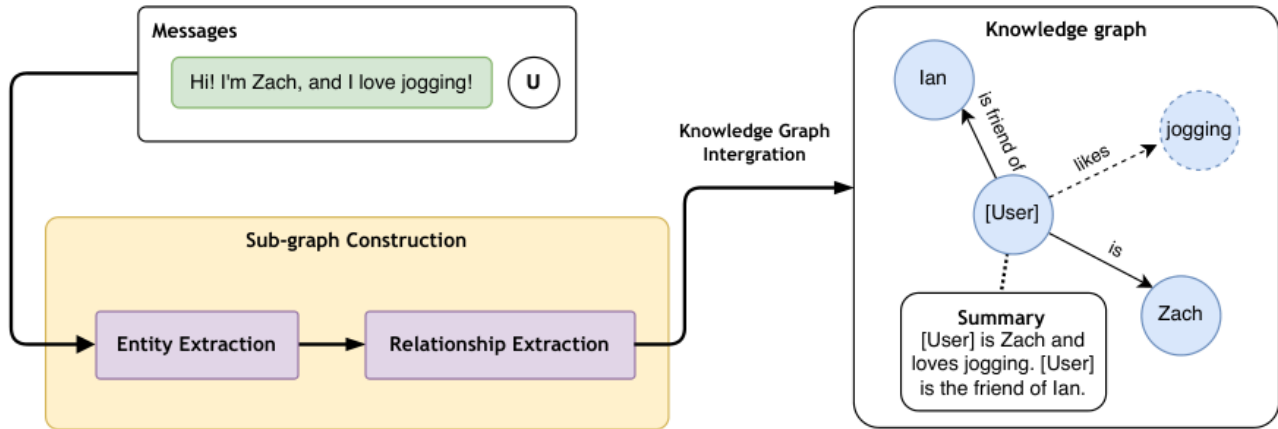


Figure 2: Overview of the memory updating pipeline. The pipeline extracts entities and relationships from the user's current message to update the knowledge graph. Entities and relationships outlined with a dotted line indicate updates.

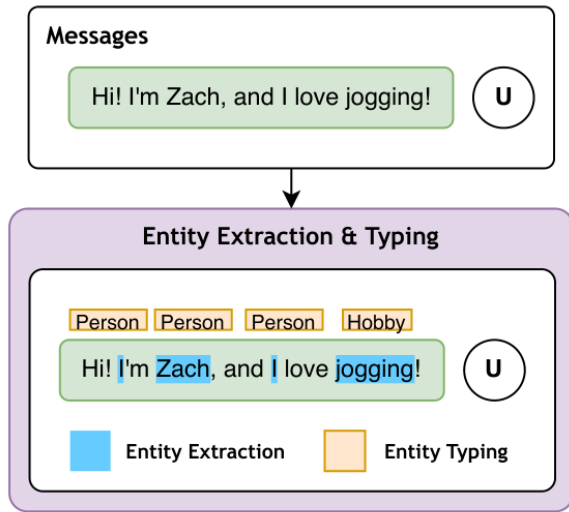


Figure 3: An illustration of the process of entity extraction and typing.

- 3.4 The path's sufficiency is evaluated by an LLM, allowing for early termination if the query is adequately addressed.
- 3.5 The loop continues, updating the current entity, until no relevant paths remain, sufficient information is found, or a step limit is reached.
- (4) **Summarisation:** A final *summary* based on the constructed *reasoning_path* and the original *query* is generated by an LLM and returned.

4 User Study

Ten participants were recruited by convenience sampling. Specific demographic data, such as age or gender, were not recorded. This

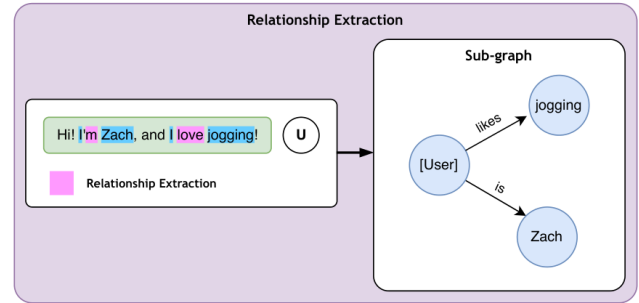


Figure 4: The process of extracting relationships between entities. The result is a knowledge subgraph representing the extracted memories.

decision reflected the pilot nature of the study and aimed to reduce participant burden, as detailed demographic analysis was not an objective of this preliminary work. However, it was noted that all participants were postgraduate students.

To protect the privacy of the participants, the conversation between them and the chatbot was not recorded and the memory was deleted at the end of the study. To ensure that the user study was conducted as expected, each participant was asked to share their screen during the study. Participants were asked in the application form about their previous experience with LLM-powered chatbots such as ChatGPT, Claude, or Gemini. In total, 9 participants had used these chatbots before the study.

The questionnaires used to measure participant perceptions of the agent's intelligence and trustworthiness are adapted from the Godspeed questionnaire [1]. Since the chatbot has no physical body, the "Anthropomorphism" and the "Animacy" sections were excluded. Furthermore, a *privacy offensiveness* criterion is added to the safety section, as the memory mechanism might make some participants feel that the agent is compromising their privacy.

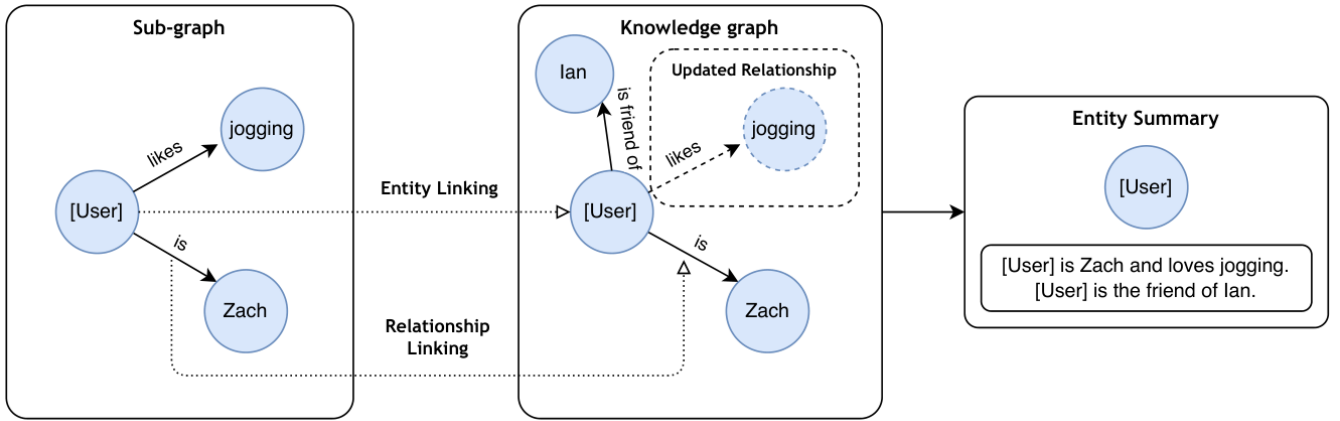


Figure 5: The pipeline for integrating the sub-knowledge graph into the main knowledge graph. The process involves linking entities in the sub-graph to the main graph, creating new entities if no matches are found, and avoiding duplicate relationships by linking existing ones. Finally, a summary is generated for each entity with updated relationships.

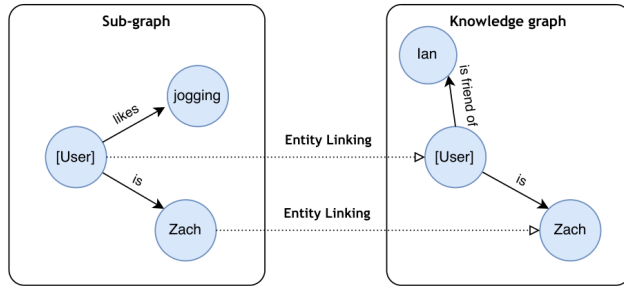


Figure 6: The entity linking process.

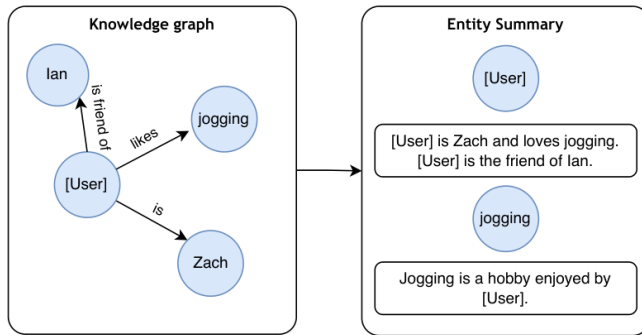


Figure 7: The summary update process.

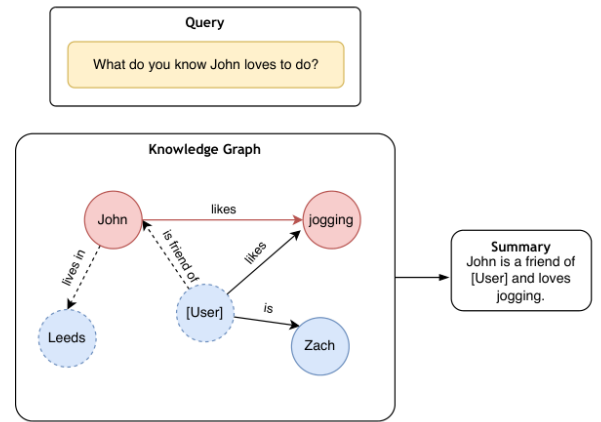


Figure 8: The query reasoning process.

- (1) Participants are exposed to the chatbot. During that step, the chatbot is designed to learn from user input but not use its memory in dialogue. Participants are not told that the chatbot has any long-term memory, only that they need to evaluate its conversational skills.
- (2) Participants are then told to fill a survey (pre-test) relating to their perception of the chatbot.

- (3) Participants are then exposed to a filler task (a guided discussion of various unrelated topics) designed to keep them busy.
- (4) Participants are then told that the session is coming to an end and that the chatbot will close the conversation. During the closing step, the chatbot is instructed to make use of what it has learnt during the conversation by recalling the name and interests of the participant, revealing that it remembers things about them.
- (5) Participants are then asked to fill a second survey (post-test).
- (6) Participants are made able to see the thinking process and the results of the memory update and retrieval, and given access to a visualisation of the chatbot's memory (illustrated in Figure 9). They are then asked to manipulate the memory through a graphical interface built on top of the graph database, and become familiar with basic memory operations (updating and retrieving memories).

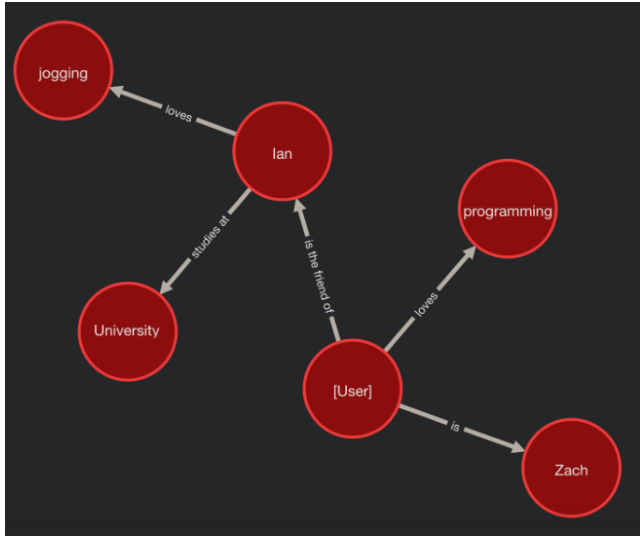


Figure 9: An example visualisation of the chatbot's memory.

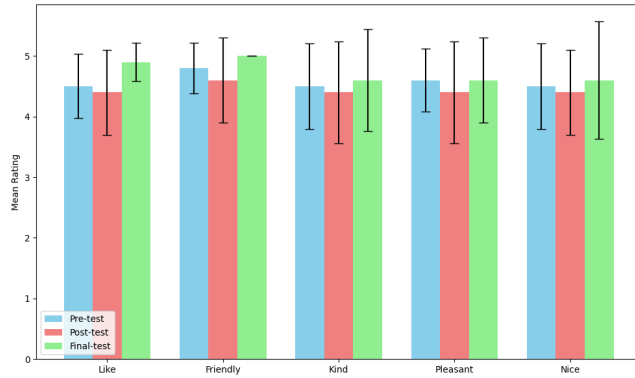


Figure 10: Mean likeability ratings across different phases of the testing process (higher is better).

- (7) Finally, participants are asked to complete the final survey (final-test).

5 Results and Discussion

Numerical results are presented in Table 1, and further illustrated through bar charts in Figures 10, 11, and 12 with respect to the chatbot's perceived likeability, intelligence, and safety (respectively).

Likeability. We observed a mixed initial reaction from participants when the chatbot demonstrated memory of previous conversations ranging from surprise and curiosity to slight discomfort. This accompanied a slight decrease in likeability scores immediately after the memory feature was introduced. However, once the underlying mechanism and information stored in the knowledge graph were revealed to the participants, their comfort levels increased, with a subsequent increase in likeability scores.

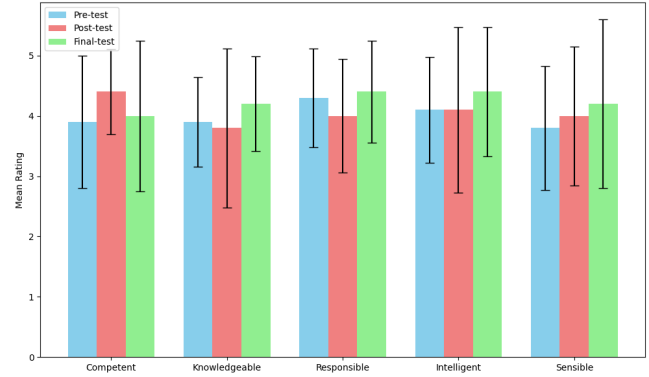


Figure 11: Mean intelligence ratings across different phases of the testing process (higher is better).

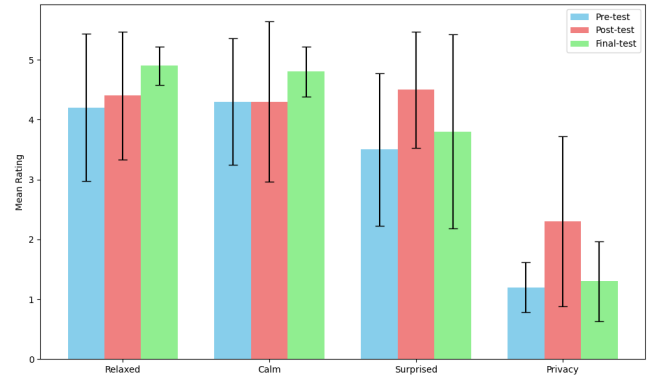


Figure 12: Mean safety ratings across different phases of the testing process (higher is better).

Perceived Intelligence. The introduction of the memory feature did not initially lead to a significant increase in perceived intelligence scores. However, after participants were presented with the chatbot's memory visualisation and a clear explanation of how the memory mechanism works, their perceptions of the chatbot's intelligence improved notably. This improvement was particularly evident in metrics related to the chatbot's knowledge, understanding, and coherence.

Perceived Safety. Participants generally felt safe and relaxed throughout the study. However, the initial surprise and uncertainty about the chatbot's memory capabilities led to a temporary increase in perceived privacy concerns. These concerns were largely mitigated once participants were provided with a clear understanding of the memory mechanism and the information stored in the knowledge graph.

6 Conclusion

In this pilot study, we investigated the impact of long-term semantic memory on the perception of LLM-powered conversational agents by users. In order to do so, we built MemoryGraph, an LLM-powered chatbot that uses a Neo4j graph database as a semantic memory

Table 1: Summary statistics of user ratings, where (↓) indicates a decrease and (↑) indicates an increase from the previous test.

	Metric	Pre-test		Post-test		Final-test	
		Mean	STD	Mean	STD	Mean	STD
Likeability	Like	4.50	0.53	4.40 (↓)	0.70	4.90 (↑)	0.32
	Friendly	4.80	0.42	4.60 (↓)	0.70	5.00 (↑)	0.00
	Kind	4.50	0.71	4.40 (↓)	0.84	4.60 (↑)	0.84
	Pleasant	4.60	0.52	4.40 (↓)	0.84	4.60 (↑)	0.70
	Nice	4.50	0.71	4.40 (↓)	0.70	4.60 (↑)	0.97
Perceived Intelligence	Competent	3.90	1.10	4.40 (↑)	0.70	4.00 (↓)	1.25
	Knowledgeable	3.90	0.74	3.80 (↓)	1.32	4.20 (↑)	0.79
	Responsible	4.30	0.82	4.00 (↓)	0.94	4.40 (↑)	0.84
	Intelligent	4.10	0.88	4.10	1.37	4.40 (↑)	1.07
	Sensible	3.80	1.03	4.00 (↑)	1.15	4.20 (↑)	1.40
Perceived Safety	Relaxed	4.20	1.23	4.40 (↑)	1.07	4.90 (↑)	0.32
	Calm	4.30	1.06	4.30	1.34	4.80 (↑)	0.42
	Surprised	3.50	1.27	4.50 (↑)	0.97	3.80 (↓)	1.62
	Privacy	1.20	0.42	2.30 (↑)	1.42	1.30 (↓)	0.67

which can be manipulated directly or through interaction with the chatbot. Although this shows promising results, there are, of course, some severe limitations to this study: (1) our sample size is both too small and too homogeneous to generalise our findings beyond the scope of the pilot; (2) because of the pilot nature of the study, we only collected quantitative data.

Our findings show that while the introduction of a knowledge graph-based memory mechanism may enhance the perceived intelligence of our conversational agent (answering our underlying research question), transparency about the mechanism’s functionality may help improve user trust, acceptance, and comfort. By providing users with clear explanations and visualisations of the memory process, we could address privacy concerns, improve likeability, and ultimately contribute to a more positive and engaging human-AI interaction experience. This sets the groundwork for future research that overcomes the limitations of this specific piece of work through a larger recruitment of a diverse pool of participants, both in demographics and in expertise with large language models; and through the collection of richer data, such as interviews, that allows us to get a more nuanced understanding of the relationship between the cognitive capabilities of conversational agents and their perceptions by users.

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References

- [1] Christoph Bartneck, Dana Kulić, Elizabeth Croft, and Susana Zoghbi. 2009. Measurement instruments for the anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety of robots. *International journal of social robotics* 1 (2009), 71–81.
- [2] Bohan Chen and Andrea L Bertozzi. 2023. AutoKG: Efficient automated knowledge graph generation for language models. In *2023 IEEE International Conference on Big Data (BigData)*. IEEE, 3117–3126.
- [3] Smit Desai, Mateusz Dubiel, and Luis A. Leiva. 2024. Examining Humanness as a Metaphor to Design Voice User Interfaces. In *Proceedings of the 6th ACM Conference on Conversational User Interfaces (Luxembourg, Luxembourg) (CUI '24)*. Association for Computing Machinery, New York, NY, USA, Article 7, 15 pages. <https://doi.org/10.1145/3640794.3665535>
- [4] Alexander Dregger, Maximilian Seifermann, and Andreas Oberweis. 2024. Language Cues for Expressing Artificial Personality: A Systematic Literature Review for Conversational Agents. In *Proceedings of the 6th ACM Conference on Conversational User Interfaces (Luxembourg, Luxembourg) (CUI '24)*. Association for Computing Machinery, New York, NY, USA, Article 22, 17 pages. <https://doi.org/10.1145/3640794.3665535>
- [5] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, et al. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783* (2024).
- [6] Melissa Guyre, Liz Holland, Nirva Shah, and Rahul R. Divekar. 2024. Prompt Engineering an LLM into Roleplaying a Management Coach: a Short Guide by and for Non-NLP Experts. In *Proceedings of the 6th ACM Conference on Conversational User Interfaces (Luxembourg, Luxembourg) (CUI '24)*. Association for Computing Machinery, New York, NY, USA, Article 49, 10 pages. <https://doi.org/10.1145/3640794.3665570>
- [7] Chenxu Hu, Jie Fu, Chenzhuang Du, Simian Luo, Junbo Zhao, and Hang Zhao. 2023. Chatdb: Augmenting llms with databases as their symbolic memory. *arXiv preprint arXiv:2306.03901* (2023).
- [8] Samuel Kernan Freire, Mina Foosherian, Chaofan Wang, and Evangelos Niforatos. 2023. Harnessing Large Language Models for Cognitive Assistants in Factories. In *Proceedings of the 5th International Conference on Conversational User Interfaces (Eindhoven, Netherlands) (CUI '23)*. Association for Computing Machinery, New York, NY, USA, Article 44, 6 pages. <https://doi.org/10.1145/3571884.3604313>
- [9] Q. Vera Liao and Jennifer Wortman Vaughan. 2023. AI Transparency in the Age of LLMs: A Human-Centered Research Roadmap. *arXiv:2306.01941 [cs.HC]* <https://arxiv.org/abs/2306.01941>
- [10] Lei Liu, Xiaoyan Yang, Yue Shen, Binbin Hu, Zhiqiang Zhang, Jinjie Gu, and Guannan Zhang. 2023. Think-in-memory: Recalling and post-thinking enable llms with long-term memory. *arXiv preprint arXiv:2311.08719* (2023).
- [11] Junru Lu, Siyu An, Mingbao Lin, Gabriele Pergola, Yulan He, Di Yin, Xing Sun, and Yunsheng Wu. 2023. Memochat: Tuning llms to use memos for consistent long-range open-domain conversation. *arXiv preprint arXiv:2308.08239* (2023).
- [12] Victor R. Martinez and James Kennedy. 2020. A Multiparty Chat-Based Dialogue System with Concurrent Conversation Tracking and Memory. In *Proceedings of the 2nd Conference on Conversational User Interfaces (Bilbao, Spain) (CUI '20)*. Association for Computing Machinery, New York, NY, USA, Article 12, 9 pages. <https://doi.org/10.1145/3405755.3406121>
- [13] Ali Modarressi, Ayyoob Imani, Mohsen Fayyaz, and Hinrich Schütze. 2023. Ret-llm: Towards a general read-write memory for large language models. *arXiv preprint arXiv:2305.14322* (2023).
- [14] Abdessalam Ouazaki, Kristoffer Bergram, Juan Carlos Farah, Denis Gillet, and Adrian Holzer. 2024. Generative AI-Enabled Conversational Interaction to Support Self-Directed Learning Experiences in Transversal Computational Thinking. In *Proceedings of the 6th ACM Conference on Conversational User Interfaces (Luxembourg, Luxembourg) (CUI '24)*. Association for Computing Machinery, New York, NY, USA, Article 13, 12 pages. <https://doi.org/10.1145/3640794.3665542>

- [15] Charles Packer, Vivian Fang, Shishir_G Patil, Kevin Lin, Sarah Wooders, and Joseph_E Gonzalez. 2023. MemGPT: Towards LLMs as Operating Systems. (2023).
- [16] Yvon Ruitenburg, Minha Lee, Wijnand IJsselsteijn, and Panos Markopoulos. 2024. Seeking Truth, Comfort, and Connection: How Conversational User Interfaces can help Couples with Dementia Manage Reality Disjunction. In *Proceedings of the 6th ACM Conference on Conversational User Interfaces* (Luxembourg, Luxembourg) (CUI '24). Association for Computing Machinery, New York, NY, USA, Article 6, 15 pages. <https://doi.org/10.1145/3640794.3665547>
- [17] Jesús Sánchez Cuadrado, Sara Pérez-Soler, Esther Guerra, and Juan De Lara. 2024. Automating the Development of Task-oriented LLM-based Chatbots. In *Proceedings of the 6th ACM Conference on Conversational User Interfaces* (Luxembourg, Luxembourg) (CUI '24). Association for Computing Machinery, New York, NY, USA, Article 11, 10 pages. <https://doi.org/10.1145/3640794.3665538>
- [18] Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, Thomas Mesnard, Bobak Shahriari, Alexandre Ramé, et al. 2024. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118* (2024).
- [19] Bing Wang, Xinnian Liang, Jian Yang, Hui Huang, Shuangzhi Wu, Peihao Wu, Lu Lu, Zejun Ma, and Zhoujun Li. 2023. Enhancing large language model with self-controlled memory framework. *arXiv preprint arXiv:2304.13343* (2023).
- [20] Shijie Zheng, Keith He, Le Yang, and Jie Xiong. 2024. MemoryRepository for AI NPC. *IEEE Access* (2024).
- [21] Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. 2024. Memorybank: Enhancing large language models with long-term memory. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 19724–19731.